Inital comments on the data and code

Ian Hussey

Study 1

Code cannot run due to missing files

Analysis code relies on "RRR_Hyperaccesibility.xlsx" and RRR_metaWE92 object, neither of which are present on OSF

The only version of RRR_Hyperaccesibility.xlsx on OSF contains simulated data.

RRR_metaWE92 object is not created or loaded by the code.

Not computationally reproducible

All three of the manuscript, final data file on OSF, and the reproduced values from the cleaned data+R code all include different estimates for the same estimand.

E.g., spot check:

Source	Site	mean age	SD age
Manuscript	Ku	20.92	2.26
RRR_WE92_all.csv	Ku	19.906	2.505
RRR_Study1.R + study 1 cleaned data xlsx files	Ku	19.946	2.526

Table of max differences for each column in RRR_WE92_all.csv vs the RRR_Study1.R + study 1 cleaned data xlsx files:

column	max_abs_diff
pct.Manipchk.pass_diff	100.000
pct.Probe.pass_diff	100.000
count.Manipchk.pass_diff	92.000
count.Probe.pass_diff	80.000
N.conc_diff	48.000
N.supp_diff	34.000
pct.N.incl_diff	26.804
N.female_diff	24.000
pct.Compliance.pass_diff	15.934
N.male_diff	7.000
max.age_diff	2.000
min.age_diff	1.000
sd.age_diff	0.620
mean.age_diff	0.375
d.high_diff	0.123
FE.GLMM.allwordtype.TP_Wordtype_diff	0.083
FE.GLMM.allwordtype.cond_TP_Wordtype_diff	0.077
d.low_diff	0.074
mean.supp.10s_diff	0.062
v.low_diff	0.059
v.high_diff	0.058
FE.GLMM.allwordtype.Wordtype_diff	0.054
FE.GLMM.allwordtype.TP_diff	0.054
FE.GLMM.allwordtype.Cond_TP_diff	0.048
sd.supp.10s_diff	0.048
FE.GLMM.allwordtype.Cond_Wordtype_diff	0.046
mean.supp.3s_diff	0.038
mean.conc.10s_diff	0.038
FE.GLMM.targetword.Cond_TP_diff	0.037
ES.nontargetwords.cond_diff	0.032
FE.GLMM.targetword.TP_diff	0.030
mean.conc.3s_diff	0.024
sd.conc.3s_diff	0.023
FE.GLMM.allwordtype.Cond_diff	0.023
sd.conc.10s_diff	0.021
sd.supp.3s_diff	0.021
VPC.m2_diff	0.021
VPC.m3_diff	0.021
ES.nontargetwords.TP_diff	0.014
FE.GLMM.targetword.Cond_diff	0.011
IRR_diff	0.000
count.Compliance.pass_diff	0.000

Differences betwen RRR_WE92_all.csv and the computationally reprodcued files can be found in RRR_Study1_reproducibility.Rmd, RRR_Study1_reproducibility.html, and results_study_1_reproducibility_diffs.csv.

Code contains errors

N.conc and N.supp were calculated incorrectly from the number of columns, not rows

```r
# original code, returns number of columns not rows
df1$N.conc = length(subset(RRR_WE92, Cond == "Concentration"))
df1$N.supp = length(subset(RRR_WE92, Cond == "Suppression"))
# corrected code
df1$N.conc = nrow(subset(RRR_WE92, Cond == "Concentration"))
df1$N.supp = nrow(subset(RRR_WE92, Cond == "Suppression"))
```

Mean and SD age for the whole sample are calculated incorrectly

code calculates the mean of the mean-age-per-site (ignoring differences in N) the SD of the mean-age-per-site (can't derive SD from means).

Original: M = 20.75, SD = 1.85

Corrected: M = 20.58, SD = 4.01

In appropriate rounding

All rounding in the code is done via round(), which does not round the way people think

`round(3.5)` = 4, but `round(4.5)` also = 4.

all round() should likely be changed to janitor::round_half_up().

Double rounding

`df1` is rounded, then summarized across sites, then rounded again for reporting. Double rounding should be avoided as it distorts estimates.

Correspondence with manuscript

Not yet checked given that the results are not yet computationally reproducible from code and data.

Study 2

Not checked yet.